

Test→ Train↓	Detector↓ Metrics→	ChatGPT		Llama-2-70b		Google-PaLM		Claude-instant		Avg.	
		AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.
ChatGPT	Roberta	98.38 $\pm$ 0.32	90.52 $\pm$ 1.93	81.71 $\pm$ 2.27	54.64 $\pm$ 16.57	74.56 $\pm$ 0.89	20.20 $\pm$ 12.87	66.74 $\pm$ 1.49	22.36 $\pm$ 13.26	80.35 $\pm$ 1.24	46.93 $\pm$ 11.16
	LRR	92.61 $\pm$ 0.39	26.20 $\pm$ 0.00	95.84 $\pm$ 0.34	86.20 $\pm$ 0.00	81.98 $\pm$ 0.14	39.80 $\pm$ 0.00	57.82 $\pm$ 0.80	0.10 $\pm$ 0.00	82.06 $\pm$ 0.42	38.07 $\pm$ 0.00
	DetectGPT	54.87 $\pm$ 0.25	0.11 $\pm$ 0.14	59.21 $\pm$ 0.53	0.66 $\pm$ 1.77	55.29 $\pm$ 0.37	0.08 $\pm$ 0.06	55.92 $\pm$ 0.61	0.00 $\pm$ 0.00	56.32 $\pm$ 0.44	0.21 $\pm$ 0.49
	Fast-Detect.	75.65 $\pm$ 0.28	11.60 $\pm$ 0.00	85.49 $\pm$ 0.31	2.50 $\pm$ 0.00	80.36 $\pm$ 0.63	17.70 $\pm$ 0.00	47.29 $\pm$ 0.44	0.00 $\pm$ 0.00	72.20 $\pm$ 0.42	7.95 $\pm$ 0.00
	Str-Detect.	52.05 $\pm$ 0.00	0.01 $\pm$ 0.00	52.30 $\pm$ 0.00	0.01 $\pm$ 0.00	56.05 $\pm$ 0.00	0.01 $\pm$ 0.00	57.50 $\pm$ 0.00	0.01 $\pm$ 0.00	54.47 $\pm$ 0.00	0.01 $\pm$ 0.00
	Binoculars	97.36 $\pm$ 0.52	40.70 $\pm$ 0.00	99.45 $\pm$ 0.44	98.70 $\pm$ 0.00	98.03 $\pm$ 0.34	89.80 $\pm$ 0.00	61.86 $\pm$ 3.64	3.40 $\pm$ 0.00	89.18 $\pm$ 1.24	58.15 $\pm$ 0.00
	RepreGuard	99.84 $\pm$ 0.12	100.00 $\pm$ 0.00	99.55 $\pm$ 0.12	99.26 $\pm$ 0.11	88.67 $\pm$ 0.65	64.66 $\pm$ 1.26	85.00 $\pm$ 1.40	56.12 $\pm$ 2.20	93.26 $\pm$ 0.57	80.01 $\pm$ 0.89
Llama-2-70b	Roberta	95.49 $\pm$ 0.96	71.16 $\pm$ 4.59	94.21 $\pm$ 2.06	63.66 $\pm$ 10.40	86.00 $\pm$ 2.17	43.60 $\pm$ 2.56	76.98 $\pm$ 4.04	22.88 $\pm$ 5.13	88.17 $\pm$ 2.31	50.32 $\pm$ 5.67
	LRR	91.88 $\pm$ 1.05	26.20 $\pm$ 0.00	96.47 $\pm$ 0.52	86.20 $\pm$ 0.00	81.65 $\pm$ 0.55	39.80 $\pm$ 0.00	56.20 $\pm$ 1.77	0.10 $\pm$ 0.00	81.55 $\pm$ 0.97	38.07 $\pm$ 0.00
	DetectGPT	54.33 $\pm$ 0.81	0.51 $\pm$ 0.80	59.36 $\pm$ 0.86	1.28 $\pm$ 2.18	55.02 $\pm$ 0.98	0.02 $\pm$ 0.06	56.11 $\pm$ 0.24	0.00 $\pm$ 0.00	56.20 $\pm$ 0.72	0.45 $\pm$ 0.76
	Fast-Detect.	94.08 $\pm$ 1.26	71.90 $\pm$ 0.00	98.76 $\pm$ 0.05	94.20 $\pm$ 0.00	92.39 $\pm$ 0.28	81.80 $\pm$ 0.00	51.45 $\pm$ 0.62	0.40 $\pm$ 0.00	84.17 $\pm$ 0.55	62.08 $\pm$ 0.00
	Str-Detect.	52.05 $\pm$ 0.00	0.01 $\pm$ 0.00	52.30 $\pm$ 0.00	0.01 $\pm$ 0.00	56.05 $\pm$ 0.00	0.01 $\pm$ 0.00	57.50 $\pm$ 0.00	0.01 $\pm$ 0.00	54.47 $\pm$ 0.00	0.01 $\pm$ 0.00
	Binoculars	97.94 $\pm$ 0.74	85.90 $\pm$ 0.00	99.63 $\pm$ 0.06	98.70 $\pm$ 0.00	97.23 $\pm$ 0.93	89.80 $\pm$ 0.00	57.49 $\pm$ 3.57	3.40 $\pm$ 0.00	88.07 $\pm$ 1.32	69.45 $\pm$ 0.00
	RepreGuard	99.54 $\pm$ 0.08	99.38 $\pm$ 0.14	99.38 $\pm$ 0.18	98.84 $\pm$ 0.11	88.84 $\pm$ 1.28	77.08 $\pm$ 0.45	84.08 $\pm$ 3.52	60.66 $\pm$ 1.66	92.96 $\pm$ 1.27	83.99 $\pm$ 0.59
Google-PaLM	Roberta	88.72 $\pm$ 1.35	40.94 $\pm$ 5.18	85.36 $\pm$ 5.00	32.70 $\pm$ 6.15	82.09 $\pm$ 3.99	36.52 $\pm$ 4.64	72.98 $\pm$ 4.00	21.54 $\pm$ 8.64	82.29 $\pm$ 3.58	32.92 $\pm$ 6.15
	LRR	91.40 $\pm$ 2.01	26.20 $\pm$ 0.00	92.84 $\pm$ 2.78	86.20 $\pm$ 0.00	83.13 $\pm$ 1.53	39.80 $\pm$ 0.00	61.11 $\pm$ 2.05	0.10 $\pm$ 0.00	82.12 $\pm$ 0.09	38.07 $\pm$ 0.00
	DetectGPT	54.04 $\pm$ 0.32	0.00 $\pm$ 0.00	59.21 $\pm$ 0.22	0.00 $\pm$ 0.00	55.30 $\pm$ 0.35	0.02 $\pm$ 0.06	55.53 $\pm$ 0.46	0.06 $\pm$ 1.63	56.02 $\pm$ 0.34	0.24 $\pm$ 0.42
	Fast-Detect.	74.62 $\pm$ 0.99	11.60 $\pm$ 0.00	86.47 $\pm$ 0.50	2.50 $\pm$ 0.00	80.66 $\pm$ 0.22	17.70 $\pm$ 0.00	48.46 $\pm$ 0.47	0.00 $\pm$ 0.00	72.55 $\pm$ 0.54	7.95 $\pm$ 0.00
	Str-Detect.	52.05 $\pm$ 0.00	0.01 $\pm$ 0.00	52.30 $\pm$ 0.00	0.01 $\pm$ 0.00	56.05 $\pm$ 0.00	0.01 $\pm$ 0.00	57.50 $\pm$ 0.00	0.01 $\pm$ 0.00	54.47 $\pm$ 0.00	0.01 $\pm$ 0.00
	Binoculars	98.56 $\pm$ 0.35	85.90 $\pm$ 0.00	99.58 $\pm$ 0.20	98.70 $\pm$ 0.00	97.98 $\pm$ 0.38	89.80 $\pm$ 0.00	61.15 $\pm$ 2.49	3.40 $\pm$ 0.00	89.32 $\pm$ 0.86	69.45 $\pm$ 0.00
	RepreGuard	98.39 $\pm$ 0.31	99.36 $\pm$ 0.07	98.53 $\pm$ 0.28	99.16 $\pm$ 0.07	93.36 $\pm$ 0.27	79.88 $\pm$ 3.43	90.57 $\pm$ 1.06	56.90 $\pm$ 2.20	95.21 $\pm$ 0.48	83.82 $\pm$ 1.44
Claude-instant	Roberta	88.63 $\pm$ 1.34	28.50 $\pm$ 11.74	77.45 $\pm$ 4.73	23.70 $\pm$ 10.19	74.90 $\pm$ 1.76	14.58 $\pm$ 13.14	88.07 $\pm$ 3.89	36.56 $\pm$ 4.83	82.26 $\pm$ 2.93	25.84 $\pm$ 9.98
	LRR	86.33 $\pm$ 4.23	26.20 $\pm$ 0.00	87.46 $\pm$ 4.43	86.20 $\pm$ 0.00	81.19 $\pm$ 3.34	39.80 $\pm$ 0.00	63.74 $\pm$ 1.07	0.10 $\pm$ 0.00	79.68 $\pm$ 3.27	38.07 $\pm$ 0.00
	DetectGPT	53.95 $\pm$ 0.84	0.02 $\pm$ 0.06	57.90 $\pm$ 1.09	0.00 $\pm$ 0.00	54.33 $\pm$ 1.13	0.06 $\pm$ 0.07	55.97 $\pm$ 0.38	0.00 $\pm$ 0.00	55.54 $\pm$ 0.86	0.02 $\pm$ 0.03
	Fast-Detect.	81.27 $\pm$ 5.40	71.90 $\pm$ 0.00	82.35 $\pm$ 4.48	94.20 $\pm$ 0.00	80.85 $\pm$ 4.70	81.80 $\pm$ 0.00	61.35 $\pm$ 0.35	0.40 $\pm$ 0.00	76.46 $\pm$ 3.73	62.08 $\pm$ 0.00
	Str-Detect.	52.05 $\pm$ 0.00	0.01 $\pm$ 0.00	52.30 $\pm$ 0.00	0.01 $\pm$ 0.00	56.05 $\pm$ 0.00	0.01 $\pm$ 0.00	57.50 $\pm$ 0.00	0.01 $\pm$ 0.00	54.47 $\pm$ 0.00	0.01 $\pm$ 0.00
	Binoculars	92.36 $\pm$ 1.95	85.90 $\pm$ 0.00	93.18 $\pm$ 1.01	98.70 $\pm$ 0.00	92.83 $\pm$ 1.18	89.80 $\pm$ 0.00	73.92 $\pm$ 0.17	3.40 $\pm$ 0.00	88.07 $\pm$ 1.08	69.45 $\pm$ 0.00
	RepreGuard	97.20 $\pm$ 1.39	96.92 $\pm$ 0.49	97.51 $\pm$ 0.99	97.16 $\pm$ 0.40	87.77 $\pm$ 2.37	63.04 $\pm$ 0.73	92.76 $\pm$ 0.49	56.22 $\pm$ 5.22	93.81 $\pm$ 1.31	78.34 $\pm$ 1.71
Multi-LLMs	Roberta	92.13 $\pm$ 2.80	60.42 $\pm$ 10.23	87.84 $\pm$ 4.06	45.90 $\pm$ 15.30	82.07 $\pm$ 3.28	31.16 $\pm$ 5.57	82.99 $\pm$ 5.42	25.64 $\pm$ 4.52	86.26 $\pm$ 3.89	40.78 $\pm$ 8.91
	LRR	92.78 $\pm$ 0.27	26.20 $\pm$ 0.00	94.98 $\pm$ 0.27	86.20 $\pm$ 0.00	81.94 $\pm$ 0.05	39.80 $\pm$ 0.00	59.93 $\pm$ 0.41	0.10 $\pm$ 0.00	82.41 $\pm$ 0.25	38.07 $\pm$ 0.00
	DetectGPT	54.92 $\pm$ 0.13	0.11 $\pm$ 0.14	59.05 $\pm$ 0.22	0.02 $\pm$ 0.06	55.29 $\pm$ 0.37	0.08 $\pm$ 0.06	55.98 $\pm$ 0.35	0.00 $\pm$ 0.00	56.31 $\pm$ 0.27	0.05 $\pm$ 0.06
	Fast-Detect.	94.98 $\pm$ 1.72	71.90 $\pm$ 0.00	95.97 $\pm$ 2.65	94.20 $\pm$ 0.00	93.19 $\pm$ 1.48	81.80 $\pm$ 0.00	56.57 $\pm$ 3.33	0.40 $\pm$ 0.00	85.18 $\pm$ 2.30	62.08 $\pm$ 0.00
	Str-Detect.	52.05 $\pm$ 0.00	0.01 $\pm$ 0.00	52.30 $\pm$ 0.00	0.01 $\pm$ 0.00	56.05 $\pm$ 0.00	0.01 $\pm$ 0.00	57.50 $\pm$ 0.00	0.01 $\pm$ 0.00	54.47 $\pm$ 0.00	0.01 $\pm$ 0.00
	Binoculars	97.95 $\pm$ 0.89	85.90 $\pm$ 0.00	98.05 $\pm$ 1.17	98.70 $\pm$ 0.00	97.58 $\pm$ 0.70	89.80 $\pm$ 0.00	68.59 $\pm$ 3.55	3.40 $\pm$ 0.00	90.54 $\pm$ 1.58	69.45 $\pm$ 0.00
	RepreGuard	98.00 $\pm$ 0.43	99.26 $\pm$ 0.11	98.44 $\pm$ 0.24	99.20 $\pm$ 0.12	92.35 $\pm$ 0.46	74.74 $\pm$ 4.52	93.40 $\pm$ 0.74	56.52 $\pm$ 1.04	95.55 $\pm$ 0.47	82.43 $\pm$ 1.45

Table 1: ID and OOD Performance Comparison of Detection Algorithms on Different Train and Eval Settings. RepreGuard shows the strongest detection performance on all settings, with an average of 96.34 $\pm$ 0.27% and 93.49 $\pm$ 1.13% AUROC (AUR.), and 83.74 $\pm$ 1.56% and 81.13 $\pm$ 2.11% TPR@0.01 (TPR.) in ID and OOD respectively. We conduct 5 rounds of bootstrapping and report the mean, standard deviation, and 95% confidence interval. Here, the subscript represents the standard deviation (e.g., 99.84 $\pm$ 0.12 indicates a mean value of 99.84 with a standard deviation of 0.12). The **blue background** or **bold** indicates the best performance and the **grey background** or **underline** indicates the second best.

generated text), while its overall performance can be unstable and may experience significant drops (e.g., 82.09 $\pm$ 3.99% AUROC on Google-PaLM generated text), and its average TPR@0.01 is 56.82 $\pm$ 5.45%. Additionally, LRR, DetectGPT, and Fast-Detect perform poorly in TPR@0.01 across most datasets, suggesting they struggle in achieving accurate LGT detection within the distribution.

OOD Performance OOD performance measures how well a model detects unseen data that differs from the training set. RepreGuard demonstrates exceptional performance in OOD scenarios, with significantly smaller drops in both AUROC and TPR@0.01 compared to other detectors. This is particularly evident in the testing on Claude-instant. For instance, when trained on Google-PaLM and tested on Claude-instant, RepreGuard achieves an AUROC of 90.57 $\pm$ 1.06% and a TPR@0.01 of 56.22 $\pm$ 5.22%. In contrast,

Binoculars experiences a drop to an AUROC of 61.15 $\pm$ 2.49% and a TPR@0.01 of 3.40 $\pm$ 0.00%, while the RoBERTa classifier drops to an AUROC of 72.98 $\pm$ 4.00% and a TPR@0.01 of 21.56 $\pm$ 8.64%. Notably, Binoculars performs particularly well on the Google-PaLM test set, highlighting its preference for certain models. Furthermore, although the RoBERTa-based classifier demonstrates relatively high AUROC performance in certain scenarios, such as achieving 95.49 $\pm$ 0.96% when trained on Llama-2-70b and tested on ChatGPT, its overall performance appears relatively unstable. The combined AUROC and TPR@0.01 are only 81.27 $\pm$ 1.33% and 41.36 $\pm$ 3.87% respectively, highlighting the instability of its performance across different scenarios.

Train→ Surrogate Model↓ Metrics→	ChatGPT		Llama-2-70b		Google-PaLM		Claude-instant		Mutil-LLMs		Avg.	
	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.
LLama-3.1-8B	<u>93.20</u>	79.70	<u>94.35</u>	81.80	<u>95.05</u>	84.80	94.80	77.10	<u>96.30</u>	81.20	94.74	80.92
LLama-3.1-8B-Instruct	90.20	<u>77.10</u>	93.85	76.50	93.40	79.40	94.75	63.10	95.75	77.20	93.59	<u>74.66</u>
Llama-3-8B	92.45	69.70	90.70	70.30	94.35	73.50	93.70	65.70	94.95	69.70	93.23	69.78
Llama-3-8B-Instruct	91.45	65.10	92.30	56.80	95.05	69.10	94.60	57.60	95.45	65.30	93.77	62.78
Llama-2-7B	87.85	66.20	92.30	<u>78.70</u>	95.40	85.20	94.45	65.70	96.85	85.20	93.37	76.20
Llama-2-7B-Instruct	90.60	65.70	90.70	80.10	92.10	45.90	<u>96.10</u>	57.50	95.95	84.00	93.09	66.64
Mistral-7B	85.95	61.80	85.35	46.80	78.70	27.80	88.65	63.90	85.85	47.80	84.90	49.62
Mistral-7B-Instruct	93.85	55.80	92.35	40.70	89.05	32.90	95.25	69.80	93.35	45.40	92.77	48.92
Falcon-7B	67.55	2.30	86.50	46.00	89.70	50.50	89.80	27.90	69.15	2.10	80.54	25.76
Falcon-7B-Instruct	74.60	0.00	92.20	65.40	90.05	67.10	92.45	<u>76.20</u>	72.60	0.00	84.38	41.74
Gemma-7B	78.50	2.90	78.20	0.60	77.70	0.50	75.30	0.00	77.55	0.00	77.45	0.80
Gemma-7B-Instruct	86.50	69.60	90.95	68.40	78.95	18.10	74.70	3.40	81.70	32.00	82.56	38.30
Gemma-2B	86.55	70.20	<u>90.80</u>	74.60	89.80	64.80	91.40	36.90	92.40	72.50	90.19	63.80
Gemma-2B-Instruct	93.35	75.30	94.70	81.10	91.35	72.10	90.00	65.80	94.35	76.70	92.75	74.20
Phi-3-Mini-4K-Instruct	85.35	36.10	89.90	23.00	89.40	2.10	89.30	1.00	91.85	3.70	89.16	13.18
Phi-2	92.80	75.80	93.55	69.20	94.85	68.40	96.10	54.50	93.40	58.80	<u>94.14</u>	65.34
GPT-J-6B	83.10	9.50	89.45	64.80	89.75	55.80	86.95	24.40	85.30	15.50	86.91	34.00
GPT-Neo-2.7B	48.90	0.10	50.00	0.00	50.00	0.10	50.00	0.10	49.75	0.00	49.73	0.06

Table 2: Performance Comparison of RepreGuard Using Different Surrogate Models on 1000 “HWT-LGT” Pairs from 4 Different LLMs. The **blue background** or **bold** indicates the best performance and the **grey background** or underline indicates the second best.

4.3 Ablation Study

Surrogate Model We evaluated the performance of RepreGuard in detection using surrogate models of different sizes and structures on 1000 random sampled “HWT-LGT” pairs from 4 different LLMs. The results in Table 2 shown LLama-3.1-8B achieved the best performance with 94.74% AUROC and 80.92% TPR@0.01, and was chosen as the example surrogate model in our paper. We find that LLMs with large sizes (7B or above), such as Llama-2-7B and Mistral-7B, consistently performed well. However, smaller models do not necessarily mean poor performance. For instance, phi-2 (2.7B) and Gemma-2B-Instruct outperformed most 7B models, achieving an AUROC of 94.14% and 92.75%, suggesting that our approach can be effectively deployed even with limited computational resources. Additionally, instruction-tuned models generally performed better than their non-instruction version, though this is not always the case (e.g., LLama-3.1-8B). Therefore, the performance of surrogate models may relate to their architecture and training data, requiring proper evaluation to determine model effectiveness.

Activation Token Ratio The activation token ratio refers to the proportion of tokens selected from end to the beginning in the samples used to collect neural activation representations for HWT and LGT. Each token’s representation is influenced by all preceding tokens in the text, making this parameter critical for optimizing LGT detec-

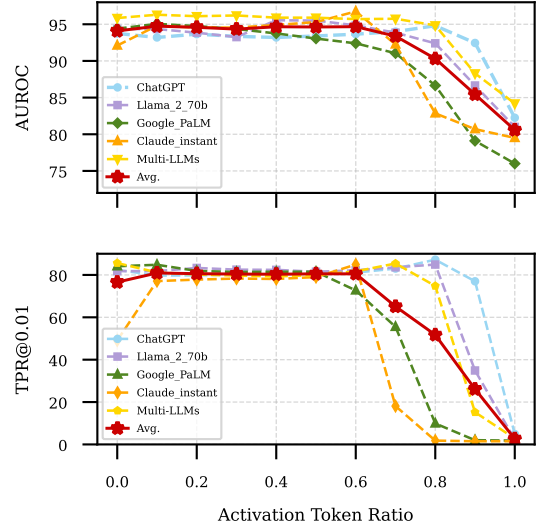


Figure 4: Effect of Activation Token Ratio in Terms of AUROC and TPR@0.01 (%) by 4 Different LLMs (250 “HWT-LGT” Pairs for Each LLM). The model name in the legend refers to the model-generated text used for representation features modeling and threshold setting.

tion. By focusing on the tokens with the most informative representations, the activation token ratio ensures a better balance between signal and noise. The results in Figure 4 indicate that the average AUROC and TPR@0.01 remains relatively stable as the activation token ratio increases, peaking at a ratio of 0.1, after which it stays consistent until approximately 0.6, followed by a sharp decline. This pattern suggests that not all token

Shots→ Train↓	Detector↓ Metrics→	16		32		64		128		256		512		1024	
		AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.	AUR.	TPR.
ChatGPT	Roberta	60.55	0.40	42.05	0.00	64.10	0.00	68.05	0.90	84.30	5.90	86.40	48.00	88.55	44.70
	LRR	80.55	31.50	81.10	31.50	82.30	31.50	82.85	31.50	83.35	31.50	83.45	31.50	82.75	31.50
	DetectGPT	53.68	0.15	54.07	0.04	54.02	0.19	55.89	0.08	56.22	0.13	57.88	0.06	57.78	0.17
	Fast-Detect.	84.80	58.80	84.85	58.80	84.75	58.80	84.70	58.80	84.75	58.80	84.70	58.80	84.70	58.80
	Binoculars	<u>84.70</u>	<u>72.60</u>	<u>84.70</u>	<u>72.60</u>	<u>88.00</u>	<u>72.60</u>	<u>88.00</u>	<u>72.60</u>	<u>90.25</u>	<u>72.60</u>	<u>88.45</u>	<u>72.60</u>	<u>90.15</u>	<u>72.60</u>
	RepreGuard	83.20	81.00	82.55	83.70	91.05	81.60	90.50	81.00	93.15	78.60	93.20	79.70	93.85	79.40
Llama-2-70b	Roberta	52.55	0.20	52.15	0.20	41.65	0.00	60.45	0.00	83.85	27.20	85.85	30.30	95.50	<u>73.40</u>
	LRR	75.40	31.50	81.10	31.50	82.80	31.50	82.95	31.50	82.45	31.50	82.45	31.50	82.80	31.50
	DetectGPT	54.12	0.11	53.89	0.07	54.33	0.18	56.01	0.02	55.76	0.16	57.23	0.09	58.09	0.13
	Fast-Detect.	79.35	58.80	<u>84.10</u>	58.80	84.20	58.80	84.50	58.80	85.00	58.80	84.70	58.80	84.70	58.80
	Binoculars	<u>83.00</u>	<u>72.60</u>	<u>83.00</u>	<u>72.60</u>	<u>86.10</u>	<u>72.60</u>	<u>86.10</u>	<u>72.60</u>	<u>87.25</u>	<u>72.60</u>	<u>87.25</u>	<u>72.60</u>	87.25	72.60
	RepreGuard	89.55	79.20	89.55	78.70	96.35	77.90	91.85	82.00	95.50	81.70	94.35	81.80	<u>94.05</u>	82.70
Google-PaLM	Roberta	55.45	0.50	61.10	0.10	66.30	0.00	71.25	2.40	77.50	2.80	<u>89.40</u>	22.90	95.45	64.50
	LRR	78.95	31.50	81.15	31.50	81.35	31.50	82.15	31.50	82.30	31.50	83.35	31.50	83.20	31.50
	DetectGPT	54.12	0.08	54.45	0.19	54.89	0.12	55.76	0.04	56.34	0.17	57.21	0.06	57.95	0.15
	Fast-Detect.	76.20	58.80	83.50	58.80	85.35	58.80	85.35	58.80	85.25	58.80	85.25	58.80	85.30	58.80
	Binoculars	<u>86.00</u>	<u>72.60</u>	<u>88.95</u>	<u>72.60</u>	<u>88.95</u>	<u>72.60</u>	<u>88.95</u>	<u>72.60</u>	88.95	<u>72.60</u>	88.45	<u>72.60</u>	89.85	<u>72.60</u>
	RepreGuard	91.70	76.30	92.00	80.40	94.25	85.30	94.95	85.10	95.05	85.50	95.05	84.80	<u>95.30</u>	82.60
Claude-instant	Roberta	51.90	0.00	62.30	0.00	50.85	0.00	62.10	0.20	74.45	9.30	<u>84.85</u>	43.90	83.70	66.00
	LRR	<u>82.50</u>	31.50	82.15	31.50	<u>83.50</u>	31.50	72.75	31.50	79.95	31.50	79.00	31.50	79.05	31.50
	DetectGPT	53.71	0.14	54.15	0.06	54.09	0.19	55.92	0.03	56.30	0.17	57.95	0.09	57.82	0.12
	Fast-Detect.	79.45	58.80	79.45	58.80	80.80	58.80	80.55	58.80	78.85	58.80	80.45	58.80	79.60	58.80
	Binoculars	82.15	72.60	<u>82.15</u>	72.60	82.15	72.60	<u>82.15</u>	<u>72.60</u>	<u>82.15</u>	<u>72.60</u>	81.90	<u>72.60</u>	<u>85.80</u>	<u>72.60</u>
	RepreGuard	92.40	<u>63.20</u>	93.25	<u>72.50</u>	94.55	<u>69.90</u>	94.60	76.30	95.40	77.00	94.80	77.10	94.75	77.30
Mutil-LLMs	Roberta	57.25	0.00	50.45	1.60	59.80	0.70	59.60	1.30	64.95	3.60	<u>93.50</u>	34.80	<u>95.65</u>	46.90
	LRR	78.25	31.50	78.25	31.50	83.30	31.50	81.90	31.50	82.35	31.50	83.50	31.50	83.50	31.50
	DetectGPT	55.48	0.04	54.56	0.02	54.43	0.15	54.97	0.18	55.93	0.01	56.31	0.03	57.11	0.07
	Fast-Detect.	<u>84.80</u>	58.80	<u>84.80</u>	58.80	85.35	58.80	85.20	58.80	85.35	58.80	85.30	58.80	85.30	58.80
	Binoculars	82.15	<u>72.60</u>	82.15	<u>72.60</u>	<u>89.30</u>	<u>72.60</u>	<u>89.30</u>	<u>72.60</u>	<u>89.45</u>	<u>72.60</u>	90.45	<u>72.60</u>	90.45	<u>72.60</u>
	RepreGuard	94.20	87.10	95.85	86.90	95.70	81.90	94.20	79.90	95.80	79.80	96.30	81.20	96.00	80.20

Table 3: Performance Comparison of RepreGuard on Various Training Data Shots in Terms of AUROC (%) on 1000 “HWT-LGT” Pairs from 4 Different LLMs. The blue background or **bold** indicates the best performance and the grey background or underline indicates the second best.

positions contribute equally to feature modeling. Specifically, the tokens at the end of the text are more distinctive in differentiating between LGT and HWT texts. As the extends toward the beginning of the sentence, noise gradually increases, leading to a decline in performance. Thus, an optimal activation token ratio is essential for balancing the modelling of important detection features while minimizing noise.

Shots of Training Dataset We evaluated the impact of the number of samples used to set the detection threshold, as shown in Table 3. The results indicate that RepreGuard is significantly more effective with limited data compared to other detectors. Specifically, RepreGuard demonstrates optimal performance in the 16-shot setting across most training datasets, achieving an average AUROC of 90.21% and TPR@0.01 of 77.36%, surpassing the best statistics-based baseline Binoculars by 6.61% and 4.76%, respectively. In addition, while the RoBERTa-based classifier achieved an average AUROC of 91.77%, which is lower than

RepreGuard’s average AUROC of 94.79% in the 1024-shot setting, its TPR@0.01 is significantly lower, averaging only 59.1%. This indicates that the RoBERTa-based classifier struggles to reliably identify positive cases under stringent false positive constraints, highlighting a limitation in its ability to balance sensitivity and specificity in such scenarios. In contrast, other detectors, such as LRR, DetectGPT and Fast-DetectGPT, demonstrate unstable performance, and consistently fail to exceed a 90% AUROC even with 1024-shot settings. These findings emphasize the strong and robust detection capabilities of RepreGuard, especially when limited data is available for training.

5 Reliability in the Wild

To evaluate our method in real-world scenarios, we conducted experiments from multiple perspectives, including performance in OOD domains, sensitivity to text length, robustness against paraphrase and perturbation attacks, and the impact of different sampling strategies.